

Although the *PCA* is implemented to spatially locate songs within a quadrant established in *Russell's Core Affect Framework* as a result of linear combinations of scaled sonic values, there is a particular subset of songs associated with a *Primary Feeling* that maps to a different *Core Affect Quadrant*. Specifically, approximately 64.4% of the songs presented in *Exploratory PCA: Mapping Song Sonics and Feelings* are associated with a *Primary Feeling* that maps to a different *Core Affect Quadrant*. As a greater number of presented songs belong to this subset, the *PCA* demonstrates that the underlying mathematics used to process sonic values do not always accurately predict the *Primary Feeling* of a respondent in response to a song. The *PCA* preparation phases exclude comprehension of human factors such as unique emotional reactivity and cultural influence. When listening to a song, individuals experience a *Primary Feeling* as a function of not only absorbed sonic qualities, but also unique emotional reactivity, and cultural influence. Subsequently, all three factors require evaluation to understand why individuals experience a *Primary Feeling* in response to a song. *Exploratory PCA: Mapping Song Sonics and Feelings* illustrates a plethora of songs that mathematically cluster to one quadrant within *Russell's Core Affect Framework*, yet are associated with a *Primary Feeling* mapping to a different *Core Affect Quadrant*.

Approximately 10.3% of the subset mathematically cluster to spatial *Quadrant 1* of *Exploratory PCA: Mapping Song Sonics and Feelings*, yet are associated with a *Primary Feeling* that maps to a different *Core Affect Quadrant*. Within the selected 10.3% of the subset, *Calm Down* by *Rema*, *Yes I'm A Mess* by *AJR*, and *Aloha Ke Akua* by *Nahko And Medicine For The People* are associated with a *Primary Feeling* mapping to *Quadrant 4*. *Quadrant 1* and *Quadrant 4* are neighbors as they each represent high

*arousal* and positive *valence*, and low *arousal* and positive *valence*. *Calm Down* by *Rema* is associated with a *Primary Feeling* of *Calm*, *Yes I'm A Mess* by *AJR* is associated with a *Primary Feeling* of *Validated*, and *Aloha Ke Akua* by *Nahko And Medicine For The People* is associated with a *Primary Feeling* of *Hopeful*. Considering the *PC2* range of the songs, the maximum *PC2* coordinate is approximately 1.504, and the minimum *PC2* coordinate is approximately 0.057. The three songs mathematically locate to the bottom half of *Quadrant 1*. As a result, the three songs are within reasonable proximity of *Quadrant 4*. While the cleaning and processing phases to prepare *PCA* data exclude understanding of why respondents experienced a *Primary Feeling*, the *PC2* range and coordinate locations may suggest that the sonic values of each song predict certain feelings that are similar to the *Primary Feeling* of the corresponding respondent when measured through *valence* and *arousal*. As for the remaining songs in the subset associated with a *Primary Feeling* that maps to a different *Core Affect Quadrant*, approximately 44.8% of the subset mathematically cluster to spatial *Quadrant 2*, approximately 20.7% of the subset mathematically cluster to spatial *Quadrant 3*, and approximately 24.1% of the subset mathematically cluster to spatial *Quadrant 4*.

Considering the complete set of songs presented in the *PCA*, approximately 68.8% of the songs are associated with a *Primary Feeling* mapping to *Core Affect Quadrant 1*. As *Quadrant 1* represents high *arousal* and positive *valence*, a compelling question emerges as well. Do most survey respondents select a favorite song as a function of experiencing high *arousal* and *valence*?